

MORGAN BAUER

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Education

University of Pennsylvania

Candidate for Master of Science in Engineering in Robotics — GRASP Laboratory

Philadelphia, PA
Expected May 2027

- **Awards & Organizations:** Climate Fellow — Climate Leaders @ Penn, Judge — UPenn EcoVenture Challenge

Eckerd College

Candidate for Bachelor of Science in Computer Science and Mathematics

St. Petersburg, FL
Conferred May 2025

- **Awards:** Edmund L. Gallizzi Award for Excellence in Computer Science, Meacham Mathematics Memorial (M^3) Award
- **Organizations:** Ford Apprentice Scholar, Pi Mu Epsilon Florida Gamma Chapter
- **Scholarships:** Natural Sciences Achievement, Music Performance

Experience

Arizona Cardinals

Training Camp Assistant — Football Analytics

Glendale, AZ
July 2026 – August 2026

- Applied machine learning techniques, including graph neural networks, variational autoencoders, and proximal policy optimization, with on-field tracking data; quantify quarterback pass IQ, predict rushing yards, and improve draft preparation.

United States Geological Survey

Information Technology Intern

St. Petersburg, FL
September 2024 – May 2025

- Served as a member of a five-person IT team assisting the St. Petersburg branch's geoscience researchers with Python and R support, software installation, and configuration.

Research & Projects

Eckerd College

Undergraduate Thesis — Deep Learning for Musical Accompaniment Generation

St. Petersburg, FL
July 2024 – May 2025

- Demonstrated the viability of using a transformer architecture for real-time musical accompaniment generation from a solo input stream.

Computer Science Research Assistant

June 2023 – May 2025

- Developing a new version of DARWIN (Digital Analysis and Recognition of Whale Images on a Network) which serves hundreds of marine scientists globally. Contributed to an updated user interface for fin tracing and contour extraction.

Ford Apprentice Scholar — Module-Lattice-Based Key-Encapsulation Mechanism (FIPS 203)

April 2023 – May 2025

- Nominated, selected, and funded to develop research and pedagogy skills. Completed additional curriculum and implemented and conducted an analysis of FIPS 203.

Undergraduate Mathematics Researcher

December 2022 – May 2025

- Researched topics in enumerative combinatorics including pop-stack-sorting and the Steiner k -Wiener index of ordered trees.

Publications & Presentations

- **Publication:** Bauer, M. and Copenhaver, K., 2025. A new lower bound for deterministic pop-stack-sorting. *European Journal of Combinatorics*, 124, p. 104046.
- **Invited Talk:** Pop-stack-sorting in a special session on enumerative combinatorics at the 55th Southeastern International Conference on Combinatorics, Graph Theory, and Computing.
- **Presentation:** Analysis and Implementation of Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203) at the 2024 Eckerd College NAS Student Research Symposium.
- **Presentation:** On the Steiner k -Wiener Index of Ordered Trees at the 2024 Eckerd College NAS Student Research Symposium.

Skills

Robotics & Systems:

- ROS, Gazebo, ARMv8 Assembly
- Git, Linux, $\LaTeX$, Feedback Control
- Kinematics, Motion Planning

Data Science & ML:

- PyTorch, TensorFlow, Keras
- NumPy, Jupyter Notebooks

Languages & App Dev:

- Python, R, SQL, Swift,
- C#, Java, XCode, Microsoft Office

Interests & Certifications

SCUBA Diving:

- TDI Technical Diver (ANDP)
- SSI Dive Guide, Rescue Diver

Other Outdoors:

- Biking, Hiking, Camping
- Snowboarding, Birding

Creative & Games:

- Woodwinds (especially Tenor Sax)
- Board Games, Reading
- Wildlife Photography